Amit Kumar *Tiwari*

DATABASE ARCHITECT | SENIOR BACKEND ENGINEER

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Professional summary

Database Architect and Senior Backend Engineer with 20+ years of hands-on experience designing, optimizing, and operating large-scale enterprise and financial systems. Deep expertise in PostgreSQL performance engineering, wallet and ledger architectures, high-volume transaction platforms, and cloud-hosted backend services on GCP and AWS. Proven ability to deliver reliable, secure, and scalable data platforms supporting production databases exceeding 450GB, complex financial workflows, and mission-critical operations.

Core technical competencies

Databases & data engineering. PostgreSQL, MySQL, SQL Server, Oracle (migration), PL/pgSQL, stored functions and triggers, indexing strategies, query optimization, schema design, replication, ETL, data validation, point-in-time recovery.

Architecture & backend. System and API architecture, transaction workflows, idempotency patterns, reliability engineering, access control, audit trails, REST APIs, Node.js, PHP/CodeIgniter.

Cloud & infrastructure. Google Cloud (Compute Engine, Cloud SQL, GKE, Cloud Run, Cloud Build, Cloud Storage, IAM, Monitoring), AWS (EC2, RDS, Aurora, App Runner, S3, SES), Docker, Linux, Nginx, PM2, CI/CD, Cloudflare, SSL/TLS.

Domain expertise. Fintech wallets and ledgers, subscription and commission engines, vesting and payout systems, blockchain-integrated workflows, fantasy sports platforms, ERP and e-governance systems.

Professional experience

Senior Software Engineer & Database Analyst

ESP Softtech Pvt. Ltd., Indore · **November 2019 – Present**

- Architect and maintain high-performance PostgreSQL and MySQL schemas, data access layers, and Node.js REST services powering transaction-intensive financial and web platforms.
- Designed and implemented end-to-end wallet balances, double-entry ledger systems, subscription lifecycles, payout/withdrawal approval workflows, audit trails, and reconciliation processes for financial products.
- Developed complex database functions, triggers, scheduled jobs, partial indexes, and role-based security controls supporting referral trees, staking, vesting, matching, and reward calculation engines.
- Engineered and operated production workloads on Google Cloud (Compute Engine, Cloud SQL, GKE, Cloud Run, Cloud Build) including IAM policies, service connectivity, automated backups, and monitoring.
- Managed AWS and Linux-based production services using Docker, Nginx, PM2, Cloudflare, and CI/CD pipelines; diagnosed and resolved deployment, networking, and dependency issues under production pressure.
- Contributed to blockchain-integrated platforms (ANANT / Anexa / StepMint) covering token subscription records, virtual rewards, wallet accounting, and secure withdrawal processing.
- Built backend systems for fantasy sports contests including scoring, leaderboards, and settlement workflows, while supporting MLM referral structures and commission payout cycles.
- Strengthened system reliability through transaction controls, idempotency patterns, access restrictions, backup strategies, and continuous SQL performance tuning.

Senior Software Engineer

Elbiz Systems Pvt. Ltd. · **September 2005 – July 2019**

- Led database architecture and engineering for large-scale enterprise ERP, e-governance, and citizen-service platforms, including production databases exceeding 450GB.
- Planned and executed complex data migrations across PostgreSQL, Oracle, SQL Server, MySQL, MS Access, and Excel sources using ETL processes, reconciliation, and data integrity validation.
- Designed reporting frameworks and backend integrations; implemented replication, backup strategies, disaster recovery planning, and SQL performance optimization for high-availability environments.
- Modernized legacy data models and applications while supporting payment processing and municipal service workflows across multiple government and enterprise clients.

Selected solution highlights

Financial platform architecture. Designed wallet and transaction ledger systems with clear payout state transitions, credit/burn flows, settlement records, and full auditability.

PostgreSQL engineering. Built complex stored functions for referral-tree traversal and reward calculations; applied partial indexes, large-table tuning techniques, and tightly scoped database role permissions.

Cloud delivery & operations. Automated build-to-VM deployment pipelines, resolved GCP IAP/SSH and networking issues, configured Nginx reverse proxies, and managed production incident response.

Data lifecycle & security. Established backup and retention policies, evaluated point-in-time recovery approaches, enforced migration validation gates, and implemented database security controls.

Education

2001 **Diploma in Computer Programming** — C-DAC Pace

2001 **Master of Commerce (M.Com)** — Devi Ahilya Vishwavidyalaya (DAVV)

1999 **Bachelor of Commerce (B.Com)** — Devi Ahilya Vishwavidyalaya (DAVV)

Additional information

Operating systems. Linux (Red Hat, CentOS, Fedora, SUSE), Windows.

Languages. English, Hindi.

Availability. Open to remote, hybrid, and on-site opportunities.